B. Tech Degree VI Semester (Supplementary)Examination September 2010

ME 602 DYNAMICS OF MACHINERY

(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART - A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Differentiate between static and dynamic force analysis.
(b) State and explain D'Alemberts principle.
(c) Derive the expression for energy stored in a flywheel.
(d) Explain the gyroscopic effects in ships.
(e) Explain static and dynamic balancing.
(f) Is complete balancing of an engine with reciprocating and rotating manner possible? Explain.
(g) Explain centrifugal tension and initial tension in belt drives.
(h) Explain belt transmission type dynamometers.

PART – B

(4 x 15 = 60)

II. The dimensions of a four bar link mechanism are AB = 500 mm, BC = 660 mm, CD = 560 mm and AD = 1000 mm. The link AB has an angular velocity of 10.5 rad/s counter clockwise and an angular retardation of 26 rad/s^2 at the instant when it makes an angle of 60° with AD, the fixed link. The masses of the links BC and CD is 4.2 Kg/m length. The link AB has a mass of 3.54 Kg, the centre of which lies at 200 mm from A and has a moment of inertia of 88500 Kg mm^2. Neglecting gravity and friction effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces. (15)

OR

III. The following data relate to a horizontal reciprocating engine Mass of reciprocating parts = 120 Kg, Crank length = 90 mm, Engine speed = 600 rpm and for the connecting rod, the Mass = 90 Kg length between centres = 450, distance of centre of mass from big end centre = 180 mm and radius of gyration about an axis through centre of mass = 150 mm. Find the magnitude and direction of the inertia torque on the crank shaft when the crank has turned 30° from the inner-dead centre. (15)

IV. The torque delivered by a two stroke engine is represented by $T = (1000 + 300\sin 2\theta - 500\cos 2\theta)\, N.m$ where θ is the angle turned by the crank from the inner dead centre. The engine speed is 250 rpm. The mass of the flywheel is 400 Kg and radius of gyration 400 mm. Determine (i) the power developed (ii) the total percentage fluctuation of speed (iii) the angular acceleration of flywheel when the crank has rotated through an angle of 60° from the inner dead centre (iv) maximum angular acceleration and retardation of the flywheel. (15)

OR

(P.T.O.)

V. Each arm of a porter governor is 250 mm long. The upper and lower arms are pivoted to links of 40 mm and 50 mm respectively from the axis of rotation. Each ball has a mass of 5 Kg and the sleeve mass is 50 Kg. The force of friction on the sleeve of the mechanism is 40 N. Determine the range of speed of the governor for extreme radii of rotation of 125 mm and 150 mm. (15)

VI. Four masses A, B, C and D are completely balanced. Masses C and D make angles of 90° and 195° respectively with B in the same sense. The rotating masses have the following properties. $m_b = 25$ Kg, $m_c = 40$ Kg, $m_d = 35$ Kg, $r_a = 150$ mm, $r_b = 200$ mm, $r_c = 100$ mm, $r_d = 180$ mm. Planes B and C are 250 mm apart. Determine (i) the mass A and its angular position (ii) the positions of planes A and D. (15)

OR

VII. The following data refer to a two cylinder uncoupled locomotive; rotating mass per cylinder = 280, reciprocating mass per cylinder = 300 Kg, distance between wheels = 1400 mm, distance between cylinder centres = 600 mm, diameter of treads of driving wheels = 1800 mm. Crank radius = 30 mm, radius of centre of balance mass = 620 mm, locomotive speed = 50 Km/hr, angle between cylinder cranks = 90°, dead load on each wheel = 3.5 tonnes. Determine (i) the balancing mass required in the planes of driving wheels if whole of the revolving and two third of the reciprocating masses are to be balanced (ii) the swaying couple (iii) the variation in the tractive force (iv) the maximum and minimum pressures on the rails (v) the maximum speed of the locomotive without lifting the wheels from the rails. (15)

VIII. (a) Obtain the conditions to be fulfilled for the maximum power transmission by belt drive. (7)

(b) A belt drive is required to transmit 10 KW from a motor running at 600 rpm. The belt is 12 mm thick and has a mass density of 0.001 g/mm^3. Safe stress in the belt is not to exceed 2.5 N/mm^2. Diameter of the driving pulley is 250 mm where as the speed of the driven pulley is 220 rpm. The two shafts are 1.25 m apart. The coefficient of friction is 0.25. Determine the width of the belt. (8)

OR

IX. Explain the following :

(i) single plate, multiplate and core clutch

(ii) block, band and internal expanding brakes. (15)

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